SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE:ITA0612 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I **K. Pujitha(192524149)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *K. Pujitha*

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labeled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Disease Diagnosis Using Bayes' Theorem, Naïve Bayes and Bayesian Belief Networks

Introduction:

Bayesian learning is a probabilistic machine learning approach used to predict diseases based on patient symptoms, medical history and laboratory test results. A hospital can develop an intelligent disease diagnosis system using Bayes' Theorem, Naïve Bayes classifiers and Bayesian Belief Networks. These methods help doctors make decisions by calculating the probability of different diseases.

1. Bayes' Theorem:

Bayes' Theorem is used to calculate the probability of a disease based on the available medical evidence.

$$P(D|E) = \frac{P(E|D) \times P(D)}{P(E)}$$

Where:

- (P(D)) = Prior probability of disease.
- (P(E|D)) = Probability of symptoms given the disease.
- (P(E)) = Total probability of the evidence.
- (P(D|E)) = Posterior probability of disease after observing evidence.

2. Model Design and Working:

1. Collect patient information such as age, weight, blood pressure, fever, blood sugar levels and family history.
2. Preprocess the medical data by removing errors and handling missing values.
3. Apply Bayes' Theorem to calculate disease probabilities.
4. Use the Naïve Bayes classifier to predict diseases based on patient symptoms.
5. Construct a Bayesian Belief Network to represent relationships between diseases, symptoms and medical history.
6. Display the predicted disease probability to help doctors make decisions.

3. Handling Uncertainty and Dependencies:

The system handles uncertainty by assigning probabilities rather than making fixed predictions. For example, if a patient has fever and cough, the system calculates the probability of different diseases.

Naïve Bayes assumes that symptoms are conditionally independent given the disease. However, in real situations, symptoms may be related to each other. Bayesian Belief Networks overcome this limitation by representing dependencies between variables using a directed acyclic graph (DAG).

4. Probability Learning and Real-World Constraints:

Probability learning helps the model estimate prior and conditional probabilities from historical patient records. As new patient information becomes available, the model can update its predictions.

However, incomplete medical records, incorrect data, privacy issues and limited training samples may affect accuracy. Doctors must validate the predictions before making treatment decisions.

Result:

The proposed probabilistic model helps hospitals predict diseases, manage uncertainty and support early diagnosis. It improves medical decision-making and provides useful assistance to doctors, but it should not replace professional medical judgment.

2. Spam Detection Using Naïve Bayes and Bayes Optimal Classifier

Introduction:

Spam detection is an important application of machine learning used to identify unwanted or harmful emails. An organization can develop an intelligent spam detection system using the Naïve Bayes classifier and Bayes Optimal Classifier. These algorithms classify emails into two categories: Spam and Not Spam.

1. Naïve Bayes Classifier:

Naïve Bayes is a supervised machine learning algorithm based on Bayes' Theorem. It calculates the probability of an email belonging to a particular category based on the words and features present in the email.

For example, words such as "Free", "Prize", "Offer" and "Win" may indicate spam emails.

The classifier uses:

$$P(C|X) = \frac{P(X|C) \times P(C)}{P(X)}$$

Where:

- (C) = Email category.
- (X) = Features or words in the email.
- (P(C)) = Prior probability.
- (P(C|X)) = Posterior probability.

2. Model Design and Working:

1. Collect a dataset containing previously labelled spam and genuine emails.
2. Preprocess the email text by removing unnecessary symbols and common words.
3. Extract important features such as keywords, sender details, links and email patterns.
4. Train the Naïve Bayes classifier using the training dataset.
5. Calculate the probability of each email being spam or genuine.
6. Classify the email based on the highest probability.
7. Evaluate the model using accuracy, precision, recall and F1-score.

3. Bayes Optimal Classifier:

The Bayes Optimal Classifier is a theoretical classification method that considers all possible hypotheses and combines their predictions according to their posterior probabilities.

Unlike a classifier that selects only one hypothesis, it averages predictions over all hypotheses. It provides the minimum possible expected classification error when the true probability distribution is known.

4. MLE and MDL Comparison:

- Maximum Likelihood Estimation (MLE): Selects parameter values that maximize the likelihood of observing the training data.
- Minimum Description Length (MDL): Selects a model by minimizing the total description length of the model and the data it explains.

MLE focuses on fitting the observed data, whereas MDL balances model complexity and goodness of fit.

5. Performance and Model Complexity:

A large training dataset generally improves classification performance. A small dataset may produce inaccurate probability estimates. A very complex model may overfit the training data, while a simple model may miss important patterns.

Naïve Bayes is fast and requires less computation, whereas the Bayes Optimal Classifier can be expensive because it considers multiple hypotheses.

Result:

The combination of Naïve Bayes and suitable feature selection provides an efficient spam detection system. It helps organizations identify unwanted emails, reduce security risks and improve email communication.

3. Customer Segmentation Using the Expectation-Maximization (EM) Algorithm

Introduction:

Customer segmentation is the process of dividing customers into different groups based on their purchasing behavior, spending habits and product

preferences. A retail company can use the Expectation-Maximization (EM) algorithm to identify hidden patterns in customer data and perform clustering.

The EM algorithm is an unsupervised machine learning technique used to estimate unknown parameters when data contains hidden or incomplete information. It is commonly used with Gaussian Mixture Models (GMM).

1. Model Design:

The retail company collects customer information such as:

- Customer age.
- Purchase frequency.
- Total spending.
- Product preferences.
- Shopping history.

The collected data is processed and given to the EM algorithm to identify different customer groups.

2. Working Procedure:

1. Collect customer purchasing data from the retail database.
2. Preprocess the data by handling missing values and normalizing numerical features.
3. Select the number of clusters and initialize the model parameters, such as mean, variance and mixing coefficients.
4. Perform the Expectation step (E-step).
5. Perform the Maximization step (M-step).
6. Repeat these steps until the model converges.
7. Analyze the final clusters and compare their likelihood values.

3. Expectation Step (E-step):

In this step, the algorithm calculates the probability that each customer belongs to a particular cluster. These probabilities are called responsibilities.

For example, a customer may have a 70% probability of belonging to a high-spending group and a 30% probability of belonging to a medium-spending group.

4. Maximization Step (M-step):

In this step, the algorithm updates the cluster parameters using the probabilities calculated in the E-step. It updates the mean, variance and mixing coefficients to improve the model.

Both steps are repeated until the likelihood converges or the improvement becomes very small.

5. Convergence and Initialization Challenges:

- The EM algorithm generally increases or maintains the likelihood at each iteration.
- Different initializations may produce different clustering results.
- Poor initialization can lead to local optima.
- Selecting too few or too many clusters may affect segmentation quality.
- Multiple random initializations and suitable cluster selection can improve stability.

6. Applications and Limitations:

The identified customer groups help companies provide personalized offers, product recommendations, discounts and targeted advertisements.

However, EM may be affected by missing values, incorrect cluster selection, outliers and unsuitable distribution assumptions. It can also require more computation for large datasets.

Result:

The EM algorithm effectively identifies hidden customer groups and estimates their underlying distributions. It helps retail companies understand customer behavior, improve marketing strategies and increase customer satisfaction.

4. Medical Image Classification Using Concept Learning and Candidate Elimination

Introduction:

Concept Learning is a machine learning approach that learns general rules from labelled training examples. A research team can use hypothesis space search and

the Candidate Elimination algorithm to classify medical images into two categories: Normal and Abnormal.

The system learns patterns from previously labelled medical images and uses the learned concepts to classify new images.

1. Hypothesis Space:

A hypothesis space is a collection of all possible rules that can be used to classify data. In medical image classification, features such as image texture, shape, brightness and detected abnormalities can be used to form hypotheses.

The algorithm searches this hypothesis space to identify hypotheses that are consistent with the training examples.

2. Candidate Elimination Algorithm:

Candidate Elimination maintains two boundaries:

- Specific Boundary (S): Represents the most specific hypotheses consistent with the training examples.
- General Boundary (G): Represents the most general hypotheses consistent with the training examples.

It updates these boundaries using positive and negative training examples.

3. Working Procedure:

1. Collect medical images labelled as Normal or Abnormal.
2. Extract useful features from the images, such as shape, texture and intensity.
3. Define a suitable hypothesis space.
4. Initialize the Specific boundary (S) and General boundary (G).
5. For positive examples, generalize the Specific boundary when necessary.
6. For negative examples, specialize the General boundary when necessary.
7. Continue updating the boundaries until all training examples are processed.
8. Test the learned hypotheses using unseen medical images.

4. Finite and Infinite Hypothesis Spaces:

A finite hypothesis space contains a limited number of possible hypotheses. It is easier to search and analyze.

An infinite hypothesis space contains unlimited possible hypotheses. It may require additional restrictions and more computational resources.

5. Sample Complexity and Mistake Bound Model:

Sample complexity refers to the number of training examples required to learn a concept with acceptable accuracy.

A larger hypothesis space generally requires more training examples to identify the correct concept.

The mistake bound model estimates the maximum number of prediction mistakes made by a learning algorithm under specific assumptions. It helps analyze the learning efficiency of different algorithms.

6. Generalization and Limitations:

Generalization means applying learned rules to unseen medical images. The system can perform well when the training examples represent real-world conditions.

However, medical images may contain noise, variations and complex patterns. Basic Candidate Elimination also assumes examples are consistent with the hypothesis space, so noisy or contradictory labels can cause problems.

Result:

Candidate Elimination helps identify hypotheses consistent with labelled examples and supports medical image classification. Its performance depends on the quality of training data, hypothesis space and feature selection. Medical experts must validate its predictions before clinical use.

5. Investment Prediction Using Bayesian Inference and Gibbs Algorithm

Introduction:

Financial markets are uncertain because stock prices, interest rates, company performance and economic conditions change continuously. A financial institution can use Bayesian inference, the Gibbs algorithm and probability learning to predict possible market outcomes and make better investment decisions.

Bayesian inference updates existing beliefs using new observations and calculates posterior probabilities. Gibbs sampling is a Markov Chain Monte Carlo (MCMC) technique used to estimate complex probability distributions.

1. Model Design:

The system collects financial information such as:

- Historical stock prices.
- Company financial reports.
- Market trends.
- Interest rates.
- Inflation rates.
- Economic indicators.

The collected data is used to estimate the probability of different investment outcomes, such as market growth, market decline and high investment risk.

2. Bayesian Inference:

Bayesian inference combines prior knowledge and observed financial data to calculate posterior probabilities.

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Where:

- $P(\theta)$ = Prior probability.
- $P(D|\theta)$ = Likelihood of observed data.
- $P(\theta|D)$ = Posterior probability.

For example, if previous market information suggests a particular stock has a good chance of growth, the system considers this prior knowledge. When new market data becomes available, it updates the probability of future growth.

3. Gibbs Algorithm:

Gibbs sampling is used when calculating the complete posterior distribution directly is difficult.

It works by repeatedly sampling one parameter at a time from its conditional probability distribution while keeping the other parameters fixed.

The steps are:

1. Initialize the model parameters.
2. Select one parameter.
3. Sample its value using its conditional probability distribution.
4. Repeat the process for the remaining parameters.
5. Continue sampling until the results become stable.
6. Use the collected samples to estimate posterior probabilities and investment risks.

4. Probability Learning and Risk Evaluation:

Probability learning helps the model estimate unknown parameters from historical and newly observed financial data.

Posterior probabilities help evaluate uncertainty and possible investment risks. For example, if the posterior probability of a market decline is high, the institution may choose a more cautious investment strategy.

Prior knowledge has a greater influence when historical data is limited. As new observations become available, the model updates its beliefs.

5. Computational Challenges and Optimization:

Gibbs sampling can require many iterations and may converge slowly. Its performance can be improved using suitable initialization, burn-in periods, convergence monitoring and efficient sampling methods.

Financial markets are also affected by unexpected events, so predictions cannot guarantee future returns.

Result:

Bayesian inference and Gibbs sampling provide a useful framework for predicting uncertain market outcomes and evaluating investment risks. The system updates its predictions as new information becomes available and supports better financial decision-making.